David Aronowitz

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Education

The Open University of Israel — Expected Graduation: November 2027

- B.Sc. in Computer Science | GPA: 92

Projects

Stock Trading Simulator & Portfolio Tracker | link

- Developed a full stack platform using Python, PostgreSQL and AWS to process simulated stock trades, allowing users to practice investing without risking real money.
- Designed a user friendly frontend using JavaScript, HTML, CSS and a Flask RESTful API to visualize users' portfolios, while also incorporating mathematical analyses using NumPy to support better investment decisions.
- Utilized Docker for containerization and AWS Lambda for automated daily balance logging to ensure offline data accuracy.

Software Team Member, First Robotics Competition

- Developed Java software while applying OOP principles, managed with Git, to drive and control a competition robot, integrating motors control over a CAN network with a computer vision pipeline, automating target tracking and enabling precise driver control.
- Worked closely within the software team, integrating feedback through weekly code reviews, and collaborated with hardware and mechanical teams to adapt software to system constraints under tight competition deadlines.

Research

Quantum Chaos research, Alpha Program, The Hebrew University of Jerusalem | link

- Used Python, NumPy, SciPy and FFT to numerically solve the Schrödinger equation and simulate the time evolution of a system modeling a black hole, investigating unmapped chaotic behaviors in quantum systems.
- Analyzed simulation data against theoretical models with Matplotlib, actively integrated expert feedback throughout the process , and presented the findings at the national Young Developers and Scientists competition.

Skills

Languages: Python, Java, C, C++, SQL, JavaScript, HTML, CSS.
Libraries/Tools: RESTful APIs, PostgreSQL, AWS, Docker, Git, Flask, Matplotlib, AI/LLM tools.
Concepts: Data Structures, Algorithms, Object Oriented Programming (OOP), System design.
Soft skills: Team player, works well under pressure, meets deadlines, able to take feedback.
Spoken Languages: Fluent English and Hebrew.